

Survey Planning → What are we planning to find?

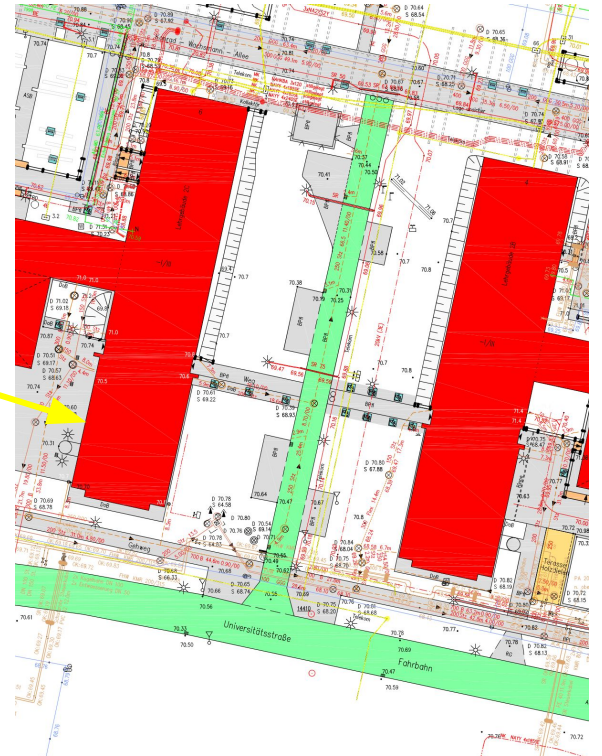


Objective

- Acquire GPR profiles along campus streets and footpaths.
- Compare results against the existing utility plan to assess what the GPR can (cannot) detect.

For each GPR profile... can we identify a reflection that matches a known utility on the plan? What does it look like in terms of depth, shape, strength?

Utility plan: location of cables, pipes, and similar infrastructure both below and above the ground.



Our equipment

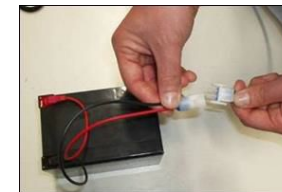
IDS Georadar: The Radar System RIS MF Hi Mod 1A consists mainly of four components:

Antenna 200 & 600 MHz

DAD control unit

Two batteries

Laptop Panasonic CF-31



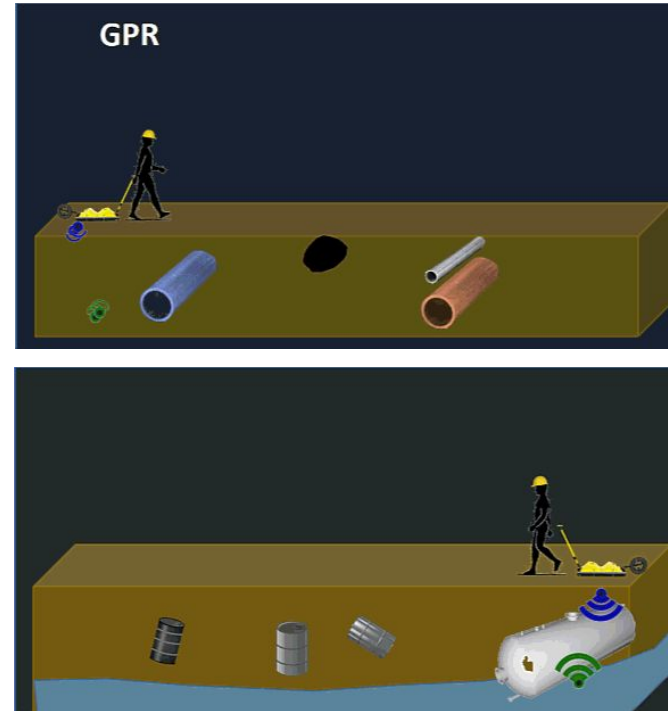
Can we find what is buried?

What should GPR detect most easily?

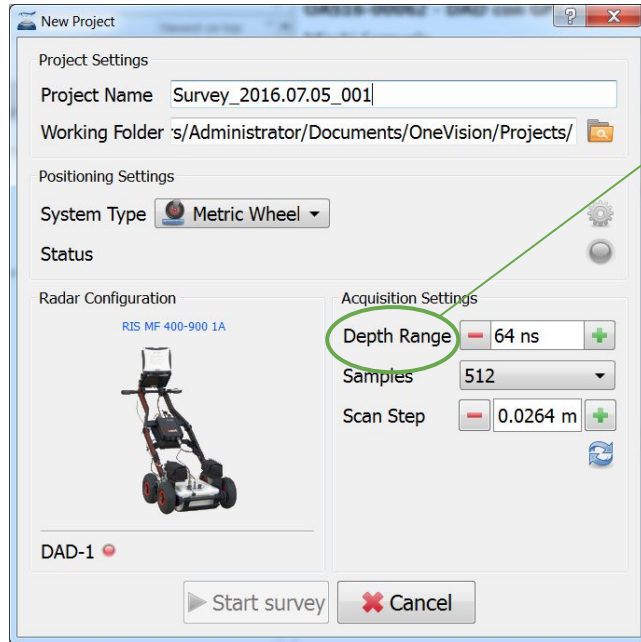
Easier: large-diameter pipes (heating, sewage) strong signal contrast, clear hyperbolic reflection.

Moderate: metallic water and gas pipes. Good reflectors if not too deep.

Harder: individual cables, small diameter, often in plastic conduits.

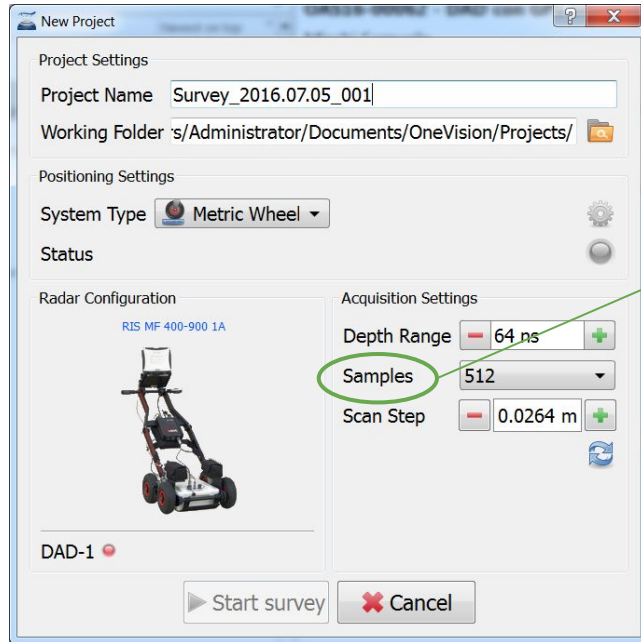


Acquisition software: important parameter setting



$$\text{propagation speed} = \frac{c}{\sqrt{\epsilon_r}}$$
$$\text{depth} = \frac{\text{velocity} * \text{time}}{2} = \frac{\text{propagation speed} * \text{range}}{2} = \frac{c}{\sqrt{\epsilon_r}} * \frac{\text{range}}{2}$$

Acquisition software: important parameter setting

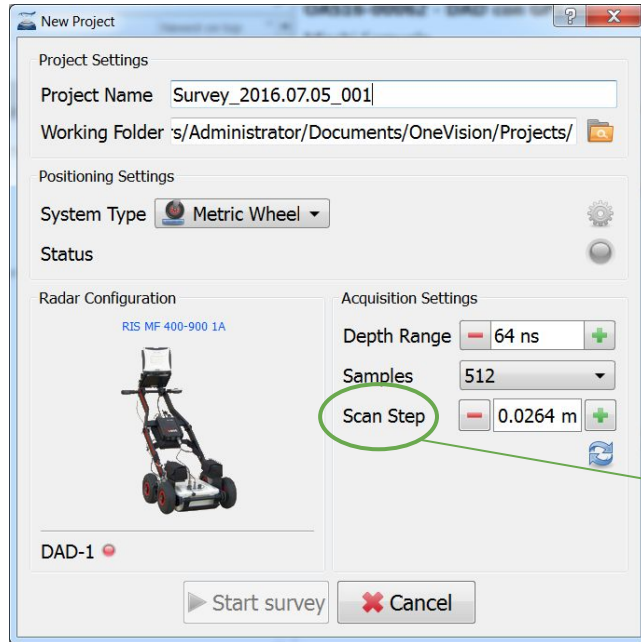


check aliasing problem →

$$\text{sampling frequency} = \frac{\text{number of samples}}{\text{range}} \gg 2 * f_{MAX} = 2 * 600\text{MHz} = 1200\text{MHz}$$

example: range=70ns, samples=512 provides sampling frequency around 7.30 GHz, which is ok

Acquisition software: important parameter setting



Lateral resolution

Careful! Electromagnetic noise

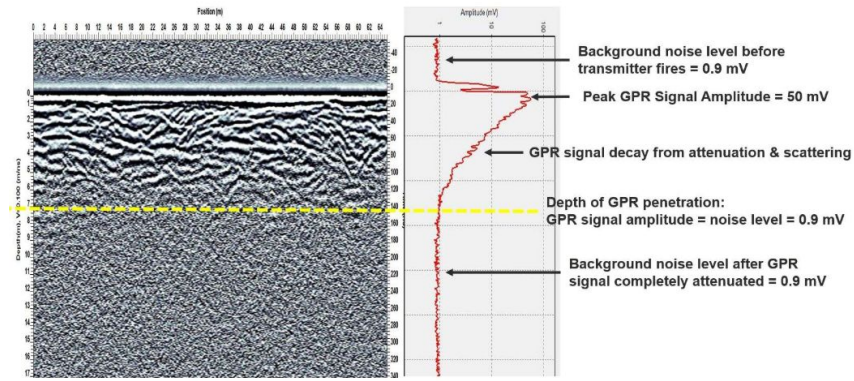


Figure 1
GPR data line (left) and Average Trace Amplitude plot (right) with a typical background noise level of 0.9 mV compared to a peak GPR signal amplitude of 50 mV.

Reflexw

The screenshot shows the website for Sandmeier geophysical research. The header includes a search bar and navigation links: Home, Reflexw, add. products, Guides, Download, how to order, about/contact, and Privacy Policy. The main content area is titled 'Welcome to - Sandmeier geophysical research -' and contains a welcome message, a description of the company's specialization in seismic and GPR processing, and a detailed list of software features. The features are categorized into GPR processing and interpretation, seismic reflection processing/interpretation, seismic refraction processing/interpretation, and borehole data processing/interpretation. A sidebar on the right contains a 'News' section with a date and a 'Full Set Software' download link. The footer includes a copyright notice for Sandmeier geophysical research 2024.

Welcome to - Sandmeier geophysical research -

We are a small geophysical company specialized in the development of software algorithms for seismic (reflection and refraction) and Ground Penetrating Radar (GPR) processing and interpretation for near surface applications.

Our software **Reflexw** is one of the world's most popular geophysical near surface processing and interpretation packages. It covers the complete range of wave data (seismic, GPR, ultrasound) and with the different geometry assemblies (surface reflection and refraction, borehole crosshole and tomography and combination of borehole and surface measurements) is therefore suitable for

- **GPR processing and interpretation**
 - import from almost all available GPR devices
 - more than 130 different processing steps including editing, filtering, migration, ...
 - interactive velocity adaptation
 - picking and layershow
 - full GPS support
- **seismic reflection processing/interpretation**
 - comprehensive prestack processing possibilities
 - interactive velocity analysis (manual interpretation, semblance, CVP, CVS, ...)
 - NMO-correction and stacking
 - prestack migration
- **seismic refraction processing/interpretation**
 - manual, semiautomatic and automatic picking
 - wavefront inversion
 - tomographic inversion using wavepath traveltimes calculation
 - forward raytracing
- **borehole data processing/interpretation**
 - for crosshole, VSP, multiangle crosshole, Walkaway VSP
 - tomographic inversion of the first arrival traveltimes
 - interactive velocity analysis of VSP-data
 - prestack migration of crosshole data

Besides this complete software package we also offer lowcost and easy to handle software solutions:

- **Reflex2DQuick**: the easy handling of 2D GPR lines or seismic shots
- **Reflex3DScan**: the fast access to a 3D-view of the underground, included within a Reflexw 3D-package
- **ReflexVibro**: interpretation of seismic vibration data

Manual and Videos →

<https://www.sandmeier-geo.de/index.html>

The screenshot shows the Reflexw software interface. The title bar reads 'Reflexw-Win'. The menu bar includes Project, Modules, Global, Help, and Exit. The 'Reflexw_Modules' menu is open, showing a list of processing modules: 2D-dataanalysis, 3D-datainterpretation, modelling, CMP-vel analysis, and traveltimes analysis. The main window displays the 'REFLEXW' logo and the text 'the 2D- and 3D-processing and interpretation package for'. Below this, there are several small images showing various geophysical data processing results, including GPR, reflection seismics, and refraction seismics. A version information box at the bottom right states: 'ReflexW/ Version 9.1.3 from 06.11.2020 copyright by K.J. Sandmeier All rights reserved academic licence number 1641 licenced to BTU Cottbus, Geopedol/Landschaftsentwicklung'.

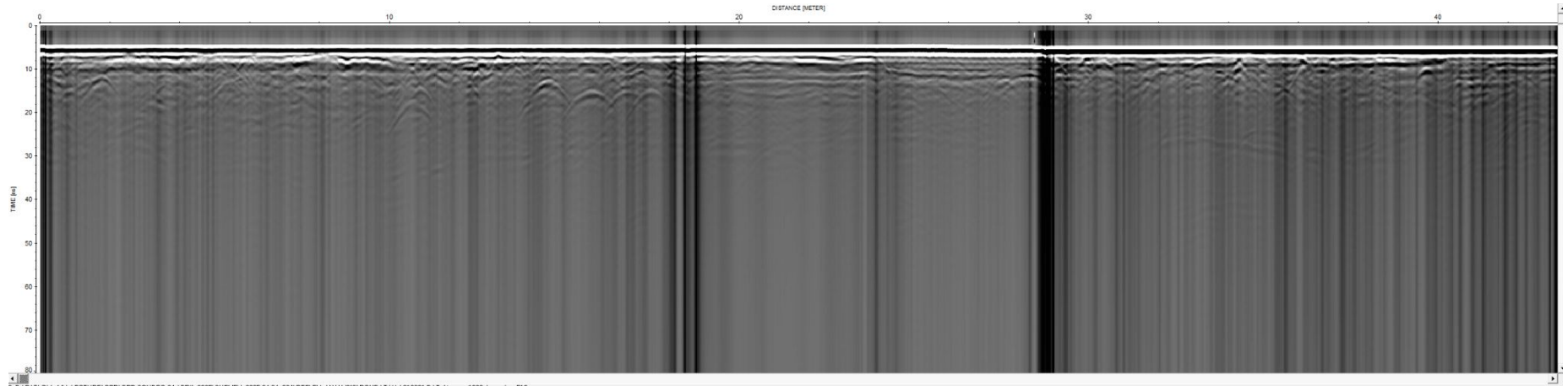
Our guide →

<https://docs.google.com/document/d/1-pYSXHyyq7VpLRZ5x-Jyaaif0vGIBCSb9d42LRsN3Jw/edit?usp=sharing>

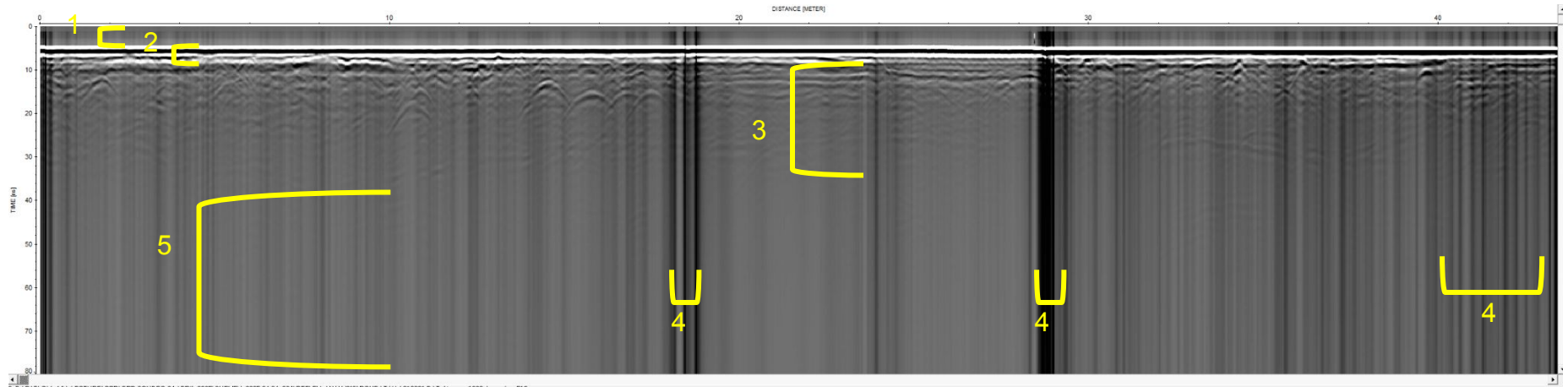
Data processing

- 1) **Get familiar with ReflexW:** basic steps include running HD Converter, creating project folders, loading data, importing profiles. Then explore the basics of *plot options*.
- 2) **Apply filters:** progressively apply processing filters to remove noise and enhance reflections.
- 3) **Initial visual analysis:** look at the profile globally, identify major features, distinguish between the two frequencies, and compare with the map generated in the field.

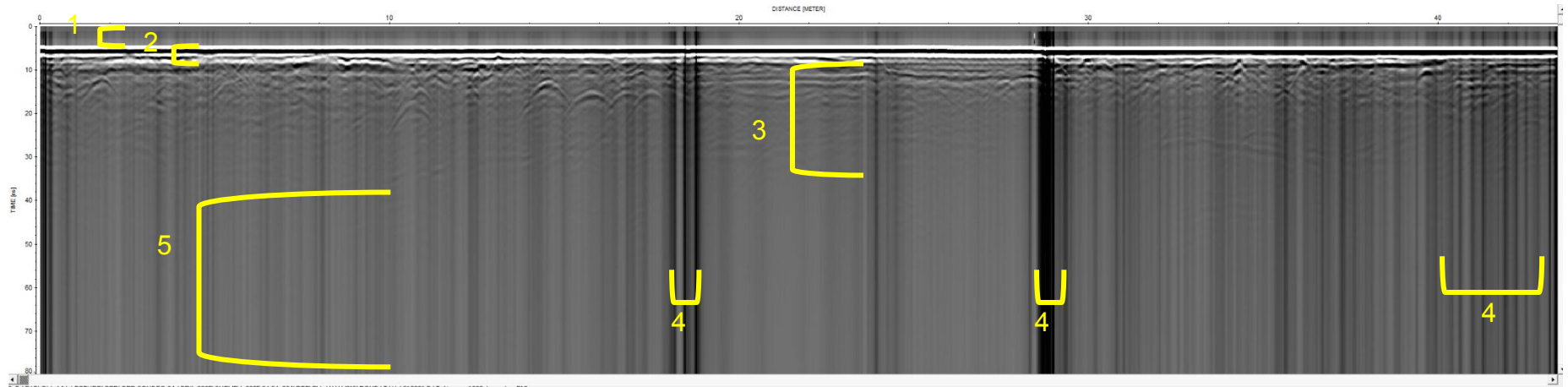
What we see from the start...



What we see from the start...

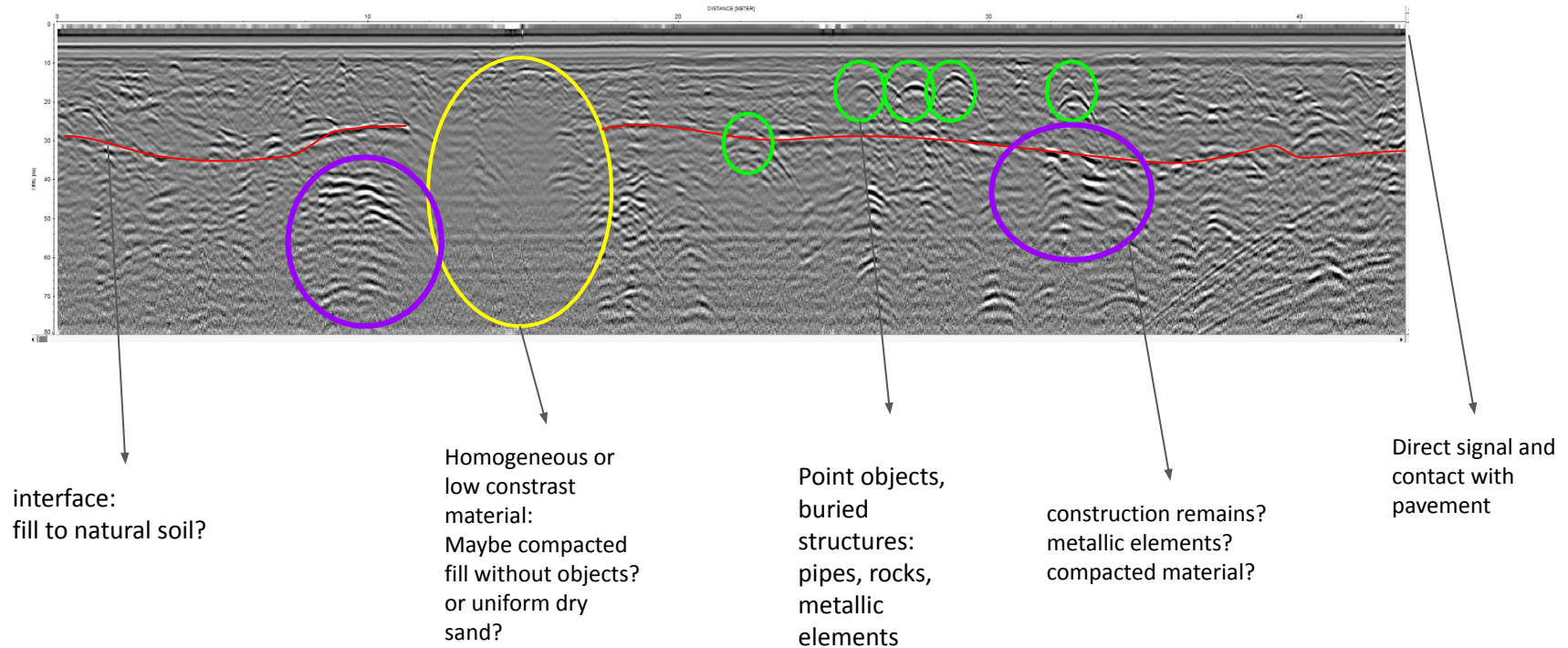


What we see from the start...



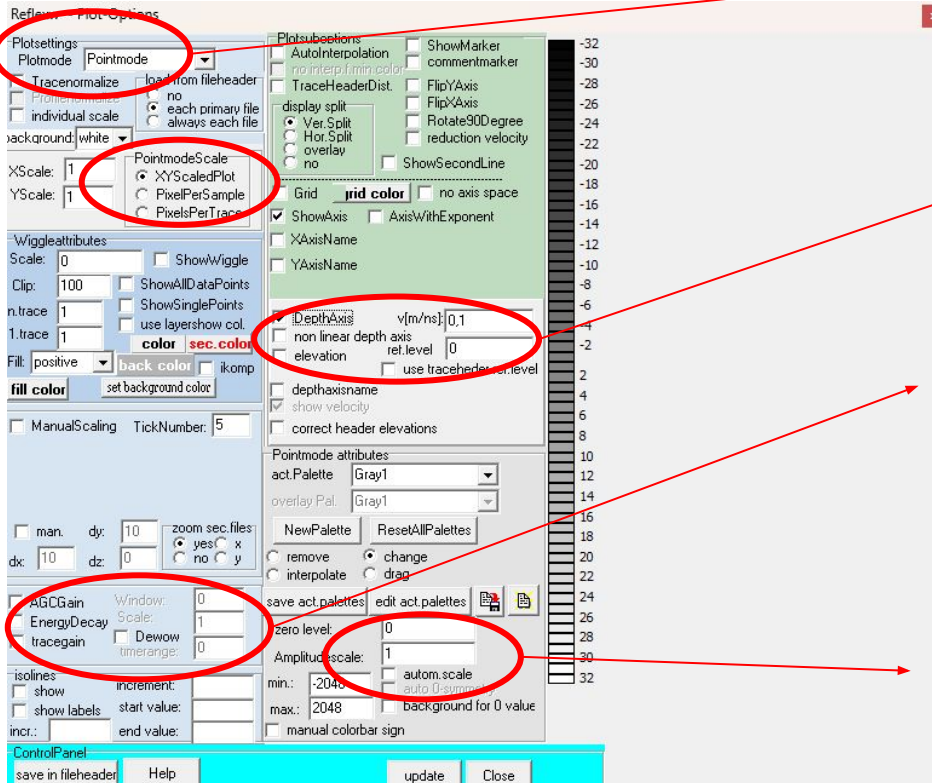
- 1) **Pre direct wave zone** → air gap or time before any signal arrives
- 2) **Direct wave** → Signal travelling directly from the transmitter to the receiver → remove with background removal or muting
- 3) **Subsurface reflections** → interfaces layers, buried objects (pipes rocks etc)
- 4) **Vertical noise** → this is caused by acquisition issues and or hardware interference. These are not real subsurface objects → spike removal
- 5) **Deep noise** → diffuse and irregular reflections; caused by signal attenuation

And after the processing...



Start working with the data.
(You can follow the guide)

Getting familiar with ReflexW:



Plotsettings:

- Choose between Wiggle mode (traces as wiggles) and Radargram mode (amplitude as grayscale).

Depth Axis:

- Set the propagation velocity (v) to convert time (ns) to depth (m). Adjusting the velocity according to the subsurface material.

Basic Filters:

- Energy Decay: compensation for the loss of signal energy with depth. We use it for visualizing deeper reflections
- Dewow: Removes low-frequency noise. Essential for getting clearer reflections.

Amplitude Scale:

- Adjusts the visual intensity of the traces : enhance weak reflections or avoid saturation

Depth Axis

- The velocity value we are going to enter in ReflexW for DepthAxis is used just to convert the vertical time axis to depth.
- If there are multiple materials, it is common to use an average value representative of the dominant material.

$$c = 30 \text{ cm/ns}$$

- Air $\epsilon_r = 1$ $v \simeq 30 \text{ cm/ns}$
- Dry Sand $\epsilon_r = 3 \text{ to } 6$ $v \simeq 12.2 \text{ cm/ns to } 17.3 \text{ cm/ns}$
- Wet Sand $\epsilon_r = 20 \text{ to } 30$ $v \simeq 5.5 \text{ cm/ns to } 6.7 \text{ cm/ns}$
- Dry Soil $\epsilon_r = 4 \text{ to } 6$ $v \simeq 12.2 \text{ cm/ns to } 15 \text{ cm/ns}$
- Wet Soil $\epsilon_r = 10 \text{ to } 15$ $v \simeq 7.7 \text{ cm/ns to } 9.5 \text{ cm/ns}$
- Fresh Water $\epsilon_r = 80$ $v \simeq 3.4 \text{ cm/ns to}$
- Concrete $\epsilon_r = 6 \text{ to } 12$ $v \simeq 8.7 \text{ cm/ns to } 12.2 \text{ cm/ns}$
- Asphalt $\epsilon_r = 4 \text{ to } 10$ $v \simeq 9.5 \text{ cm/ns to } 15 \text{ cm/ns}$

$$\text{propagation speed} = \frac{c}{\sqrt{\epsilon_r}}$$

$$\text{depth} = \frac{\text{velocity} * \text{time}}{2} = \frac{\text{propagation speed} * \text{range}}{2} = \frac{c}{\sqrt{\epsilon_r}} * \frac{\text{range}}{2}$$

Setting an incorrect speed results in depth errors:

- Too high speed: Depth is overestimated
- Too low speed: Depth is underestimated

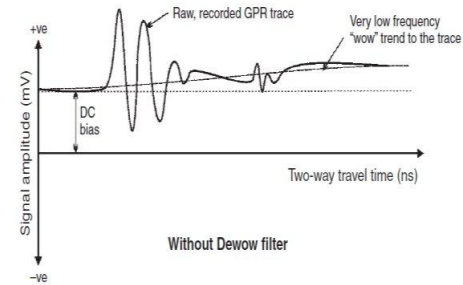
Dewow Filter: Removing Low-Frequency Noise

The filter is used to eliminate very low-frequency components that are not part of the actual reflections.

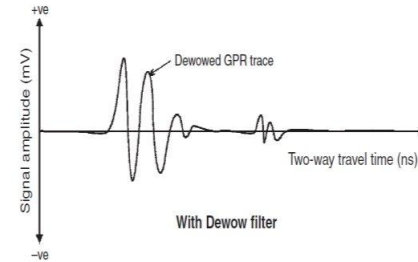
Why do we need to filter

1. **Noise from the equipment (DC noise):**
 - Electronic systems generate slow baseline drift and low-frequency noise.
 - These components appear as long, smooth oscillations or a curved baseline in the radargram.
2. **Low-frequency components within the pulse spectrum:**
 - A 200 MHz antenna does not emit a pure 200 MHz tone but a broadband pulse.
 - The emitted signal contains spectral components both below and above 200 MHz.
 - Dewow may remove very low-frequency parts of the pulse spectrum, and it won't affect useful signals if the window size is appropriate.

$$f_c = \frac{1}{2T}$$



Without Dewow filter

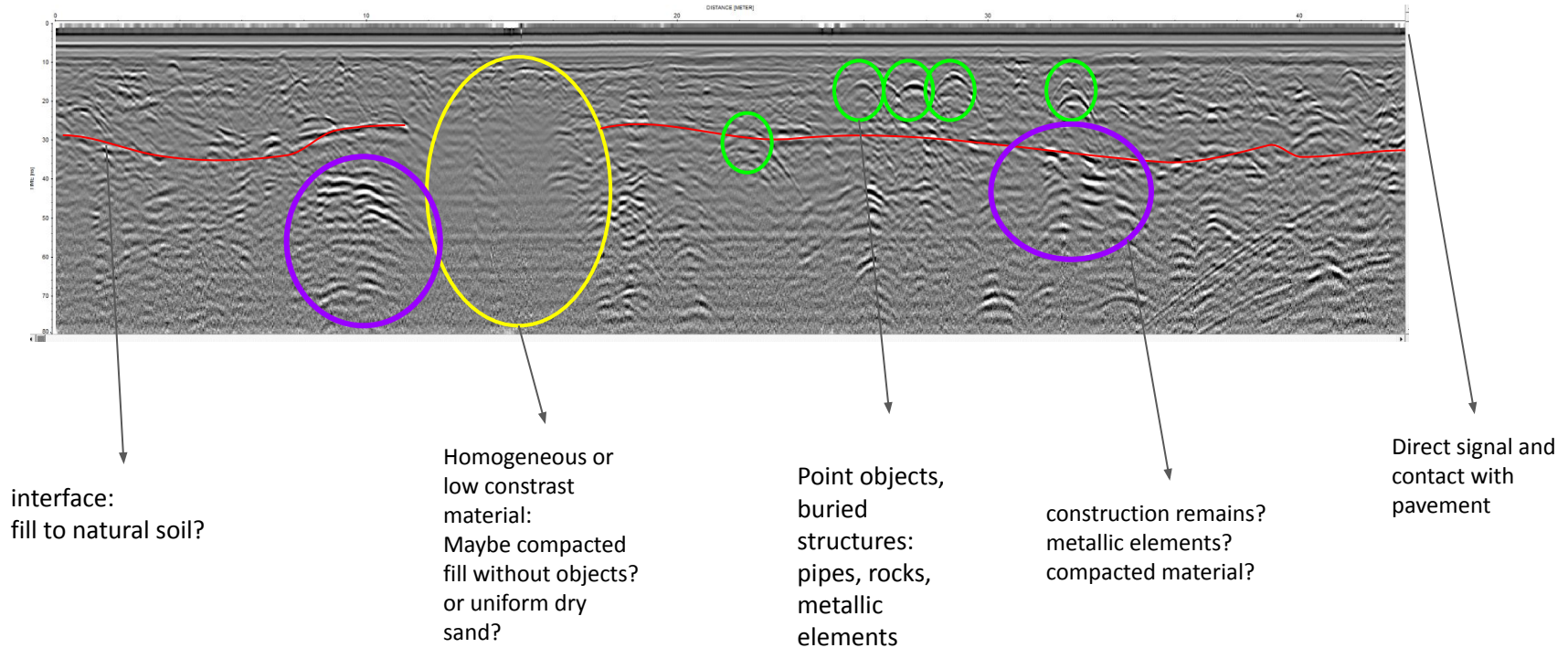


With Dewow filter

Antenna freq	Dewow time window (aprox)	Cut off freq (aprox)
200 MHz	10-20 ns	50-25 MHz
600 MHz	2-5 ns	250-100 MHz

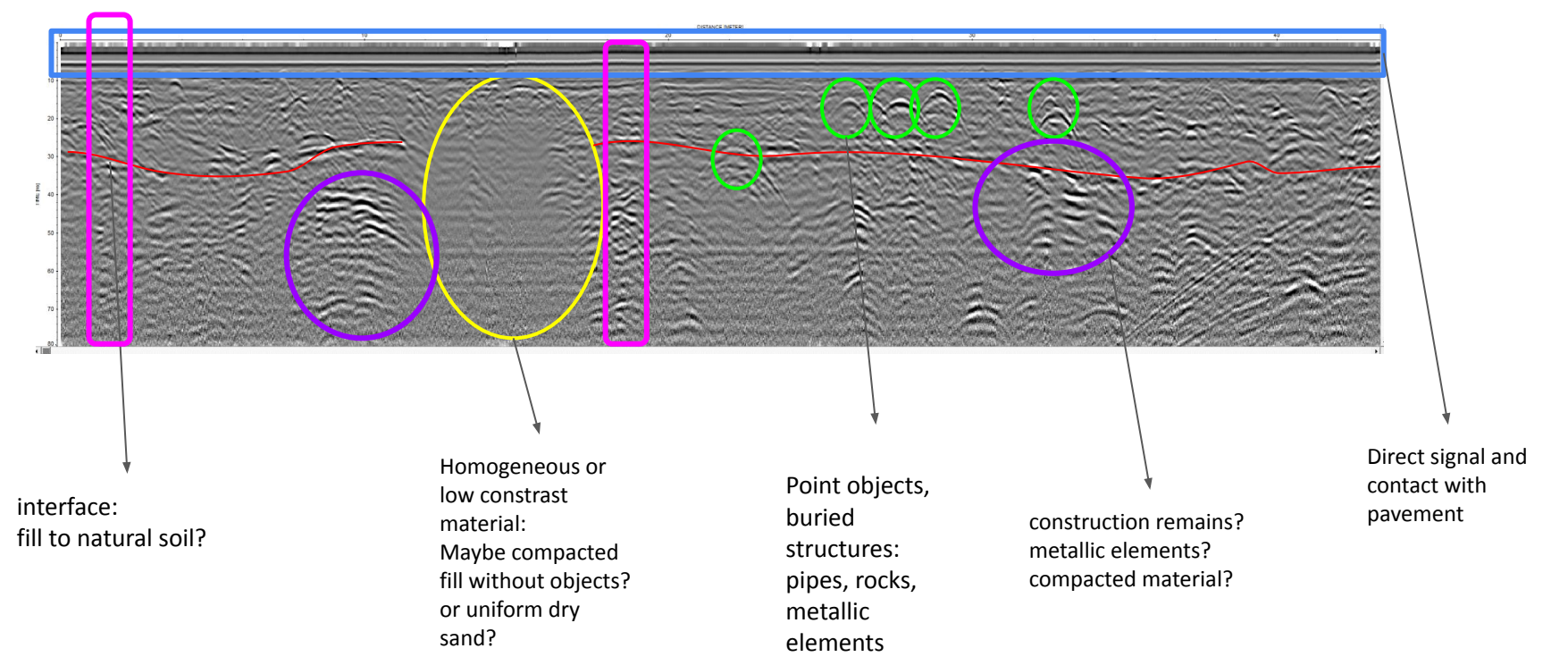
Interpretation

600 Mhz antenna - just after modifying dewow, gain, and plot scale



Interpretation

600 Mhz antenna - just after modifying dewow, gain, and plot scale



Data processing

Data import and initial setup

- Load raw data and set correct parameters (units, time window)

Basic processing

- Flip (in case is necessary)
- Static correction: Align the time zero to account for signal delay

Amplitude and gain adjustments

- Amplitude scale. You can adjust it for optimal visual contrast
- Energy decay. This compensates for amplitude loss

Noise reduction and filtering

- Background removal. This subtracts average trace to eliminate systematic noise
- Band pass filter. Keep only the frequencies you want within a useful range.

Advanced processing

- Migration. This corrects hyperbolic reflection to their true position
- Deconvolution. Increase resolution by compressing the wavelet.

DO NOT overprocess the data!